

1. Recall the fact: if $Z \sim N_p(0, I_p)$ is independent of $V \sim W_p(n-1, I_p)$, then $Z^T \left(\frac{V}{n-1}\right)^{-1} Z \sim$ **Question 1:**

- (a) Suppose we have independent samples: $\mathbf{x}_1, \dots, \mathbf{x}_n \stackrel{i.i.d.}{\sim} N_p(\mu, \Sigma)$. Let $\bar{\mathbf{x}}$ be the sample mean and $\mathbf{S}$ be the sample covariance matrix. Show that for any $C \in \mathbb{R}^{m \times p}$ satisfying $C\Sigma C^T$ is invertible, we have

$$n[C(\bar{\mathbf{x}} - \mu)]^T [C\Sigma C^T]^{-1} [C(\bar{\mathbf{x}} - \mu)] \sim \frac{(n-1)m}{n-m} F_{m, n-m}$$

- (b) Suppose we have samples from two independent populations: $\mathbf{x}_{11}, \dots, \mathbf{x}_{1n_1} \stackrel{i.i.d.}{\sim} N_p(\mu_1, \Sigma)$, $\mathbf{x}_{21}, \dots, \mathbf{x}_{2n_2} \stackrel{i.i.d.}{\sim} N_p(\mu_2, \Sigma)$. Show that

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2))^T \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{\text{pooled}} \right]^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2)) \sim \frac{(n_1 + n_2 - 2)p}{n_1 + n_2 - p - 1} F_{p, n_1 + n_2 - p - 1},$$

where $\bar{\mathbf{x}}_1$ and $\bar{\mathbf{x}}_2$ are the sample mean for $\{\mathbf{x}_{1i}\}$ and $\{\mathbf{x}_{2i}\}$.

(a) $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \sim N_p(\mu, \frac{1}{n} \Sigma)$, $S = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T \Rightarrow (n-1)S \sim W_p(n-1, \Sigma)$ and $\bar{\mathbf{x}}$ and S independent
Since $\bar{\mathbf{x}} - \mu \sim N_p(0, \frac{1}{n} \Sigma)$ $\Rightarrow \bar{\mathbf{x}} - \mu \sim N_m(0, C(\frac{1}{n} \Sigma)C^T) = N_m(0, \frac{1}{n} C\Sigma C^T)$ where $Z = C(\bar{\mathbf{x}} - \mu)$

Since $(n-1)S \sim W_p(n-1, \Sigma) \Rightarrow (n-1)C\Sigma C^T \sim W_m(n-1, C\Sigma C^T)$

Now we define $U = \frac{1}{\sqrt{n}} (C\Sigma C^T)^{-\frac{1}{2}} Z$ then $E[U] = 0$ and $\text{Cov}(U) = \frac{1}{n} (C\Sigma C^T)^{-\frac{1}{2}} (\frac{1}{n} C\Sigma C^T) (\frac{1}{n} C\Sigma C^T)^{-\frac{1}{2}} = I_m$

So $U \sim N_m(0, I_m)$ and $Z = \frac{1}{\sqrt{n}} (C\Sigma C^T)^{\frac{1}{2}} U$

$\therefore T^2 = n \bar{\mathbf{x}}^T (C\Sigma C^T)^{-1} \bar{\mathbf{x}} = U^T (C\Sigma C^T)^{-\frac{1}{2}} (C\Sigma C^T)^{\frac{1}{2}} U$. Since $(n-1)C\Sigma C^T \sim W_m(n-1, C\Sigma C^T) \Rightarrow$

$(n-1)(C\Sigma C^T)^{-\frac{1}{2}} C\Sigma C^T (C\Sigma C^T)^{-\frac{1}{2}} \sim W_m(n-1, I_m) \Rightarrow$ set $V = (n-1)(C\Sigma C^T)^{-\frac{1}{2}} C\Sigma C^T (C\Sigma C^T)^{-\frac{1}{2}} \sim W_m(n-1, I_m)$

Thus we have $T^2 = U^T (C\Sigma C^T)^{-\frac{1}{2}} (C\Sigma C^T)^{\frac{1}{2}} U = U^T \left(\frac{V}{n-1} \right)^{-1} U$ and $U \sim N_m(0, I_m)$ and $V \sim W_m(n-1, I_m)$

Thus by fact above we have $T^2 \sim \frac{(n-1)m}{n-m} F_{m, n-m}$ shown. $\square$

(b) $\bar{\mathbf{x}}_1 \sim N_p(\mu_1, \frac{1}{n_1} \Sigma)$, $\bar{\mathbf{x}}_2 \sim N_p(\mu_2, \frac{1}{n_2} \Sigma) \Rightarrow \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2) \sim N_p(0, \frac{1}{n_1} \Sigma + \frac{1}{n_2} \Sigma)$. $S_{\text{pooled}} = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2}$, $(n_1-1)S_1 \sim W_p(n_1-1, \Sigma)$, $(n_2-1)S_2 \sim W_p(n_2-1, \Sigma) \Rightarrow (n_1+n_2-2)S_{\text{pooled}} \sim W_p(n_1+n_2-2, \Sigma)$

if $Z = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2 - (\mu_1 - \mu_2)$, then $T^2 = Z^T \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{\text{pooled}} \right]^{-1} Z$. Now take $U = \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^{-\frac{1}{2}} Z$ so $U \sim N_p(0, I_p)$

Then $T^2 = U^T \left[\frac{1}{n_1+n_2-2} (n_1+n_2-2) S_{\text{pooled}} \right]^{-1} U$ with $V = (n_1+n_2-2) S_{\text{pooled}} \sim W_p(n_1+n_2-2, \Sigma)$ By similarly argument above

$T^2 = U^T \left(\frac{V}{n_1+n_2-2} \right)^{-1} U \sim \frac{(n_1+n_2-2)p}{n_1+n_2-1-p} F_{p, n_1+n_2-1-p}$ $\square$.

2. Samples of sizes $n_1 = 45$ and $n_2 = 55$ were taken of Wisconsin homeowners with and without air conditioning, respectively (assume both populations are multivariate normal and have the same population covariance matrix). Two measurements of electrical usage (in kilowatt hours) were considered. The first is a measure of total on-peak consumption (X_1) during July, and the second is a measure of total off-peak consumption (X_2) during July. The resulting summary statistics are

$$\bar{\mathbf{x}}_1 = \begin{bmatrix} 204.4 \\ 556.6 \end{bmatrix}, \quad \mathbf{S}_1 = \begin{bmatrix} 13825.3 & 23823.4 \\ 23823.4 & 73107.4 \end{bmatrix}, \quad n_1 = 45$$

$$\bar{\mathbf{x}}_2 = \begin{bmatrix} 130.0 \\ 355.0 \end{bmatrix}, \quad \mathbf{S}_2 = \begin{bmatrix} 8632.0 & 19616.7 \\ 19616.7 & 55964.5 \end{bmatrix}, \quad n_2 = 55$$

- (a). What is the pooled estimate of the population covariance matrix?
(b). Test H_0 : The two populations have equal means at the significance level 0.05.
(c). Find 95% Bonferroni intervals for the differences in the mean components.

(a) $S_{\text{pooled}} = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2} = \frac{44S_1 + 54S_2}{98} = \begin{bmatrix} 10863.6735 & 21505.4224 \\ 21505.4224 & 63661.3122 \end{bmatrix}$

(b). Test $H_0: \mu_1 = \mu_2$ at $\alpha = 0.05$:

$$T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^T S_{\text{pooled}}^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) \approx 15.66$$

$$\text{And } F = \frac{98.2}{97} F_{2, 97} = \frac{186}{97} \cdot 3.04 = 6.244$$

$$\therefore T^2 = 15.66 > 6.244 = F$$

$\therefore$ we reject H_0 , $\mu_1 \neq \mu_2$.

(c). when $k=1$: $S_{11} = 10863.6735$ $t_{n_1+n_2-2}(\frac{\alpha}{2p}) = t_{98}(0.0125) = 2.365$

when $k=2$: $S_{22} = 63661.3122$

$$\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = \sqrt{\frac{1}{45} + \frac{1}{55}} = \frac{2\sqrt{11}}{33}$$

$$\therefore \text{when } k=1: (\bar{x}_{11} - \bar{x}_{21}) \pm t_{n_1+n_2-2} \left(\frac{\alpha}{2p} \right) \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{kk}} = 74.4 \pm 2.365 \times \frac{2\sqrt{11}}{33} \sqrt{10863.6735} \Rightarrow \text{Interval: } [24.66, 124.14]$$

$$\therefore \text{when } k=2: (\bar{x}_{12} - \bar{x}_{22}) \pm t_{n_1+n_2-2} \left(\frac{\alpha}{2p} \right) \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{kk}} = 201.6 \pm 2.365 \times \frac{2\sqrt{11}}{33} \sqrt{63661.3122} \Rightarrow \text{Interval: } [81.67, 324.53]$$

3. Two variables are observed from 3 treatments. Sample data are as follows:

Treatment 1: (6, 7), (5, 9), (8, 6), (4, 9), (7, 9);

Treatment 2: (3, 3), (1, 6), (2, 3);

Treatment 3: (2, 3), (5, 1), (3, 1), (2, 3).

Some useful statistics are:

Treatment 1: $n_1 = 5$, $\bar{\mathbf{x}}_1 = (6, 8)^T$, and

$$S_1 = \begin{pmatrix} 5/2 & -3/2 \\ -3/2 & 2 \end{pmatrix}$$

Treatment 2: $n_2 = 3$, $\bar{\mathbf{x}}_2 = (2, 4)^T$, and

$$S_2 = \begin{pmatrix} 1 & -3/2 \\ -3/2 & 3 \end{pmatrix}$$

Treatment 3: $n_3 = 4$, $\bar{\mathbf{x}}_3 = (3, 2)^T$, and

$$S_3 = \begin{pmatrix} 2 & -4/3 \\ -4/3 & 4/3 \end{pmatrix}$$

- (a) Construct the MANOVA table, like Page 30 of Chapter 5 (part 1) slides. (Do not need to include the Wilk's lambda statistic Λ^* in the table)
(b) Find the Wilk's lambda statistic, then test $H_0: \tau_1 = \tau_2 = \tau_3 = 0$ (no treatment effect) at significance level $\alpha = 0.01$.

Question 3: (a) $\bar{\mathbf{x}} = \frac{5\bar{\mathbf{x}}_1 + 3\bar{\mathbf{x}}_2 + 4\bar{\mathbf{x}}_3}{12} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$

$$B = \sum_{h=1}^3 n_h (\bar{\mathbf{x}}_h - \bar{\mathbf{x}})(\bar{\mathbf{x}}_h - \bar{\mathbf{x}})^T = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} -2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \end{bmatrix} + 4 \begin{bmatrix} -1 & -3 \\ -3 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} 36 & 48 \\ 48 & 84 \end{bmatrix}$$

$$W = \sum_{h=1}^3 (n_h - 1) S_h = \begin{bmatrix} 18 & -13 \\ -13 & 18 \end{bmatrix}$$

$$T = B + W = \begin{bmatrix} 54 & 35 \\ 35 & 102 \end{bmatrix}$$

$$(b) \Lambda^* = \frac{|W|}{|T|} = \frac{135}{428} \approx 0.315 \quad \text{for } p=2, q=3, n=12$$

$$\left(\frac{n-p-1}{p} \right) \left(\frac{1-\Lambda^*}{\Lambda^*} \right) \sim F_{2p, 2(n-p-1)} = F_{4, 18}$$

$$\therefore \frac{12-2-1}{2} \cdot \left(\frac{1-0.315}{0.315} \right) \approx 11.18 > 4.77 = F_{4, 18} \therefore \text{Reject } H_0.$$

$\square$.